

The Complete Pythagorean Compendium

Volume II: Advanced Mathematical Proofs, Cosmic Geometry, and Harmonics

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Abstract

The Pythagorean system represents the first unified field theory of Western history. This expanded Volume II formally synthesizes their philosophical metaphysics, discrete number theory, continuous geometry, harmonic music ratios, and introduces the formal proofs of incommensurability, solutions to Pell's Equation, and the elemental mapping of the regular polyhedra.

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1 Metaphysics and the Quadrivium

The school operated under the foundational axiom "*All is Number.*" The pursuit of knowledge was formalized into the Quadrivium:

- **Arithmetic:** Discrete quantity at rest (Number Theory).
- **Music (Harmonics):** Discrete quantity in motion (Ratio & Proportion).
- **Geometry:** Continuous magnitude at rest (Space).
- **Astronomy (Spherics):** Continuous magnitude in motion (Cosmic Order).

1.1 The Table of Opposites (Systychia)

Reality was defined through ten fundamental, binary dualities:

1. Limit / Unlimited
2. Odd / Even
3. One / Many
4. Right / Left
5. Male / Female
6. Rest / Motion
7. Straight / Curved
8. Light / Darkness
9. Good / Evil
10. Square / Oblong

1.2 Cosmology and Units of Reality

The cosmological model introduced the **Central Fire** (the gravitational center) and the **An-tichthon** (Counter-Earth) to balance the cosmos to the perfect number 10.

- **Monad (1):** Generator of all numbers; Reason; Stability.
- **Dyad (2):** Indefinite; Opinion; Matter.
- **Triad (3):** The first true number (beginning, middle, end).
- **Tetrad (4):** Justice; The four elements; Squaring of accounts.
- **Decad (10):** Divine perfection; Completion.

2 Number Theory and Figurate Mathematics

2.1 Figurate Numbers and the Gnomon

A *gnomon* is the shape added to a geometric figure to produce a new figure of the same type.

- **Square Numbers** ($S_n = n^2$): The gnomon is the n -th odd number ($2n + 1$).
- **Triangular Numbers** ($T_n = \frac{n(n+1)}{2}$): Representing the sum of integers from 1 to n . A foundational identity is $T_n + T_{n-1} = n^2$.
- **Oblong Numbers** ($n(n + 1)$): The sum of the first n even numbers.

2.2 Number Relations and the Classical Means

The relationships between the Arithmetic (A), Geometric (G), and Harmonic (H) means of two numbers a, b :

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b} \implies G^2 = AH \quad (1)$$

3 Geometric Formalisms and Cosmic Solids

3.1 Core Geometric Theorems

- **Pythagorean Theorem:** $a^2 + b^2 = c^2$.
- **Reciprocal Theorem:** For altitude h to the hypotenuse: $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{h^2}$.
- **Internal Angles:** $\angle A + \angle B + \angle C = 180^\circ$.

3.2 Cosmic Geometry: The Regular Polyhedra

The Pythagoreans were the first to theorize the five regular polyhedra (later known as the Platonic Solids) and mapped them to the fundamental elements of the universe based on their geometric properties.

Polyhedron	Faces (F)	Vertices (V)	Edges (E)	Elemental Association
Tetrahedron	4 (Triangles)	4	6	Fire (Sharp, piercing)
Hexahedron (Cube)	6 (Squares)	8	12	Earth (Stable, solid)
Octahedron	8 (Triangles)	6	12	Air (Mobile, light)
Icosahedron	20 (Triangles)	12	30	Water (Fluid, rolling)
Dodecahedron	12 (Pentagons)	20	30	Aether/Cosmos (The Universe)

4 Proofs and Computational Algorithms

4.1 The Crisis of Incommensurability ($\sqrt{2}$)

The discovery that the diagonal of a square cannot be expressed as a ratio of two integers (p/q) shattered the Pythagorean worldview. The formal proof by contradiction is as follows:

1. Assume $\sqrt{2}$ is rational: $\sqrt{2} = \frac{p}{q}$, where p and q are coprime integers.
2. Squaring both sides yields: $2 = \frac{p^2}{q^2} \implies p^2 = 2q^2$.
3. Therefore, p^2 is even, which means p must be even. Let $p = 2k$.
4. Substitute back: $(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2$.
5. Therefore, q^2 is even, which means q must be even.
6. **Contradiction:** If both p and q are even, they share a common factor of 2, violating the assumption that they are coprime. Thus, $\sqrt{2}$ is irrational.

4.2 Approximation of Irrationals and Pell's Equation

To handle the irrational $\sqrt{2}$, the "Side and Diagonal" recursive sequence was invented:

$$S_{n+1} = S_n + D_n, \quad D_{n+1} = 2S_n + D_n \quad (2)$$

The ratio D_n/S_n converges to $\sqrt{2}$. This algorithm mathematically satisfies a specific instance of **Pell's Equation**:

$$D_n^2 - 2S_n^2 = (-1)^n \quad (3)$$

For iteration 3 ($S = 5, D = 7$): $7^2 - 2(5^2) = 49 - 50 = -1$.

4.3 Generating Pythagorean Triples

Euclid's General Formula: $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$

Conditions for Primitive Triples: $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod{2}$.

5 Advanced Harmonic Theory

By dividing a single string (monochord), consonance was defined through superparticular ratios ($n + 1 : n$).

5.1 The Diatonic Breakdown

The Pythagoreans discovered that a Whole Tone (9:8) could not be divided into two equal rational halves. Instead, it was divided into two unequal semitones:

- **Limma (Minor Semitone):** The difference between a Perfect Fourth (4:3) and two Whole Tones $((9/8)^2 = 81/64)$.

$$\frac{4/3}{81/64} = \frac{256}{243} \approx 1.0535$$

- **Apotome (Major Semitone):** The remaining portion of a Whole Tone after a Limma is removed.

$$\frac{9/8}{256/243} = \frac{2187}{2048} \approx 1.0678$$

The Pythagorean Comma: The structural discrepancy when stacking 12 perfect fifths vs. 7 octaves, which is also the exact difference between an Apotome and a Limma:

$$\text{Comma} = \frac{2187/2048}{256/243} = \frac{531441}{524288} \approx 1.013643 \quad (4)$$

6 Data Table: First 100 Primitive Pythagorean Triples

Computed via Euclid's algorithm ($a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$). Ordered symmetrically.

Table 1: Primitive Triples 1-100

No.	a	b	c	No.	a	b	c
1	3	4	5	2	5	12	13
3	8	15	17	4	7	24	25
5	20	21	29	6	12	35	37
7	9	40	41	8	28	45	53
9	11	60	61	10	16	63	65
11	33	56	65	12	48	55	73
13	13	84	85	14	36	77	85
15	39	80	89	16	65	72	97
17	20	99	101	18	15	112	113
19	60	91	109	20	17	144	145
21	24	143	145	22	88	105	137
23	51	140	149	24	19	180	181
25	80	153	173	26	119	120	169
27	69	260	269	28	21	220	221
29	104	153	185	30	23	264	265
31	96	247	265	32	25	312	313
33	32	255	257	34	160	231	281
35	204	253	325	36	28	195	197
37	27	364	365	38	76	357	365
39	29	420	421	40	132	475	493
41	31	480	481	42	240	161	289
43	115	252	277	44	140	171	221
45	33	544	545	46	252	275	373
47	35	612	613	48	37	684	685
49	39	760	761	50	41	840	841
51	43	924	925	52	204	357	405
53	45	1012	1013	54	47	1104	1105
55	49	1200	1201	56	51	1300	1301
57	53	1404	1405	58	55	1512	1513
59	57	1624	1625	60	59	1740	1741
61	61	1860	1861	62	63	1984	1985
63	65	2112	2113	64	67	2244	2245
65	69	2380	2381	66	71	2520	2521
67	73	2664	2665	68	75	2812	2813
69	77	2964	2965	70	79	3120	3121
71	81	3280	3281	72	83	3444	3445
73	85	3612	3613	74	87	3784	3785
75	89	3960	3961	76	91	4140	4141
77	93	4324	4325	78	95	4512	4513
79	97	4704	4705	80	99	4900	4901
81	101	5100	5101	82	103	5304	5305
83	105	5512	5513	84	107	5724	5725
85	109	5940	5941	86	111	6160	6161
87	113	6384	6385	88	115	6612	6613

Table 1: Primitive Triples 1-100

No.	a	b	c	No.	a	b	c
89	117	6844	6845	90	119	7080	7081
91	121	7320	7321	92	123	7564	7565
93	125	7812	7813	94	127	8064	8065
95	129	8320	8321	96	131	8580	8581
97	133	8844	8845	98	135	9112	9113
99	137	9384	9385	100	139	9660	9661